

Complete Elliptic Reduction and Two-Phase Return for a Hamiltonian Resonant Triad

Route-A source-local certificate HCS-C344

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Abstract

For the canonical complex resonant triad we prove global Liouville integrability and reduce every intensity orbit to a real cubic. Every regular orbit is written in Jacobi form with its exact intensity period. Two complete elliptic integrals of the third kind reconstruct the surviving torus phases and give the necessary and sufficient full-state return test. The zero-Hamiltonian transfer, its factor-two period, the equal-action separatrix, and maximal-Hamiltonian relative equilibria are closed separately.

1 Canonical convention and first integrals

On $\mathbb{C}^3$ use

$$\{f, g\} = -i \sum_{j=1}^3 (f_{z_j} g_{\bar{z}_j} - f_{\bar{z}_j} g_{z_j})$$

and the real Hamiltonian

$$H = z_1 z_2 \bar{z}_3 + \bar{z}_1 \bar{z}_2 z_3. \quad (1)$$

Hamilton's equations are

$$i\dot{z}_1 = \bar{z}_2 z_3, \quad i\dot{z}_2 = \bar{z}_1 z_3, \quad i\dot{z}_3 = z_1 z_2. \quad (2)$$

Writing $I_j = |z_j|^2$ and $w = z_1 z_2 \bar{z}_3$ gives

$$\dot{I}_1 = \dot{I}_2 = -2 \operatorname{Im} w, \quad \dot{I}_3 = 2 \operatorname{Im} w.$$

Consequently

$$N_1 = I_1 + I_3, \quad N_2 = I_2 + I_3, \quad H \quad (3)$$

are conserved. Direct evaluation of the displayed bracket gives pairwise involution. On the open set $I_1 I_2 I_3 \sin \Psi \neq 0$, where $\Psi = \arg z_1 + \arg z_2 - \arg z_3$, the first two differentials are independent action differentials and

$$\partial_\Psi H = -2\sqrt{I_1 I_2 I_3} \sin \Psi \neq 0.$$

Thus the three integrals are generically independent. Their common levels are bounded because $0 \leq I_3 \leq \min(N_1, N_2)$ and $I_j = N_j - I_3$ for $j = 1, 2$. Polynomial local existence and boundedness give a global flow on all of $\mathbb{R}$.

2 The real cubic and every regular intensity

Put $x = I_3$. Since $\dot{x} = 2 \operatorname{Im} w$ and $|w|^2 = x(N_1 - x)(N_2 - x)$,

$$\dot{x}^2 = 4x(N_1 - x)(N_2 - x) - H^2 =: P(x). \quad (4)$$

Let $N_- := \min(N_1, N_2)$ and $N_+ := \max(N_1, N_2)$. The function $f(x) = x(N_1 - x)(N_2 - x)$ has one maximum on $[0, N_-]$, at

$$x_* = \frac{N_1 + N_2 - \sqrt{N_1^2 - N_1 N_2 + N_2^2}}{3}, \quad H_{\max} = 2\sqrt{f(x_*)}. \quad (5)$$

For $0 < |H| < H_{\max}$, signs at 0, the maximum, N_- , N_+ , and infinity prove that P has exactly three roots

$$0 < r_1 < r_2 < N_- \leq N_+ < r_3. \quad (6)$$

They obey

$$\sum r_j = N_1 + N_2, \quad \sum_{j < k} r_j r_k = N_1 N_2, \quad r_1 r_2 r_3 = \frac{H^2}{4}. \quad (7)$$

Theorem 1 (regular elliptic orbit). *Let $0 < |H| < H_{\max}$ and define*

$$m = \frac{r_2 - r_1}{r_3 - r_1}, \quad u = \sqrt{r_3 - r_1}(t - t_0).$$

Every intensity orbit on this level is

$$x(t) = r_1 + (r_2 - r_1) \operatorname{sn}^2(u \mid m), \quad T_x = \frac{2K(m)}{\sqrt{r_3 - r_1}}. \quad (8)$$

Proof. On $[r_1, r_2]$ the cubic is $4(x - r_1)(r_2 - x)(r_3 - x)$. Substitute (8) and use $(\operatorname{sn}')^2 = (1 - \operatorname{sn}^2)(1 - m \operatorname{sn}^2)$. The resulting identity is exact. Time translation selects either initial sign of $\dot{x}$, and sn^2 has least real period $2K(m)$. $\square$

The analytic root argument and substitution are the *cubic elliptic owner* of revision round zero.

3 Two phases, not one, control full return

When $H \neq 0$, no I_j can vanish. Writing $z_j = \sqrt{I_j} e^{i\phi_j}$, division of (2) by z_j gives

$$\dot{\phi}_1 = -\frac{H}{2(N_1 - x)}, \quad \dot{\phi}_2 = -\frac{H}{2(N_2 - x)}, \quad \dot{\phi}_3 = -\frac{H}{2x}. \quad (9)$$

With $\Pi(n \mid m)$ denoting the complete third-kind integral, integration over one intensity period yields

$$\Delta_j = -\frac{H}{\sqrt{r_3 - r_1}(N_j - r_1)} \Pi\left(\frac{r_2 - r_1}{N_j - r_1} \mid m\right), \quad j = 1, 2. \quad (10)$$

Indeed, the integral over $0 \leq u \leq 2K$ is twice its complete-quarter integral. After T_x , both x and $\dot{x}$ return; together with H they determine $e^{i\Psi}$. Hence $\Delta_3 = \Delta_1 + \Delta_2$ modulo 2π .

Because a return of the nonconstant full state must occur at an integer multiple of its least intensity period, the full complex orbit is periodic if and only if

$$\frac{\Delta_1}{2\pi} \in \mathbb{Q}, \quad \frac{\Delta_2}{2\pi} \in \mathbb{Q}. \quad (11)$$

One scalar period therefore does not settle full-state recurrence.

4 Zero Hamiltonian and double-root boundaries

Suppose first that $0 < N_1 < N_2$. With $A = \sqrt{N_1}$, $B = \sqrt{N_2}$, $m = A^2/B^2$, and $u = B(t - t_0)$, every non-equilibrium orbit, up to the two phase symmetries, is

$$\begin{aligned} z_1 &= A \operatorname{cn}(u \mid m) e^{i\alpha}, & z_2 &= B \operatorname{dn}(u \mid m) e^{i\beta}, \\ z_3 &= -iA \operatorname{sn}(u \mid m) e^{i(\alpha+\beta)}. \end{aligned} \quad (12)$$

The other ordering exchanges modes 1 and 2. The Jacobi derivative identities verify (2) directly, including both amplitude-zero crossings. Intensities have period $2K(m)/B$. Translation by $2K$ changes the signs of cn and sn but not dn , so the full state has least period $4K(m)/B$.

At $N_1 = N_2 = N > 0$, the limit $m = 1$ gives

$$\sqrt{N} (\operatorname{sech} u e^{i\alpha}, \operatorname{sech} u e^{i\beta}, -i \tanh u e^{i(\alpha+\beta)}), \quad u = \sqrt{N}(t - t_0). \quad (13)$$

It is heteroclinic between opposite points on the z_3 -axis and has infinite period.

At $|H| = H_{\max}$, the two accessible roots coalesce at x_* . The intensities are constant and the phases have frequencies

$$\omega_1 = -\frac{H}{2(N_1 - x_*)}, \quad \omega_2 = -\frac{H}{2(N_2 - x_*)}, \quad \omega_3 = -\frac{H}{2x_*} = \omega_1 + \omega_2. \quad (14)$$

The final equality is equivalent to $f'(x_*) = 0$. This relative equilibrium is periodic exactly when ω_1/ω_2 is rational. The symmetric face is always periodic; the asymmetric example $(N_1, N_2) = (5, 8)$ has $x_* = 2$, $H_{\max} = 12$, and absolute frequencies $(2, 1, 3)$.

The origin and each complex coordinate axis are equilibrium families. Introducing a real coupling g multiplies the vector field by g : the $g = 0$ face is the identity flow, and $g \neq 0$ is the present theorem after $\tau = gt$. Complex conjugation paired with time reversal leaves all intensities unchanged. These clauses form the *two-phase return owner* added in revision round one.

Source boundary

This paper is a source-local reconstruction, not a priority claim. Manley and Rowe own their energy relations; Armstrong et al. own the coupled optical amplitude setting and explicit three-wave solutions; Kaup, Reiman, and Bers give an authoritative primary treatment of broader resonant three-wave dynamics. The exact software receipts are local implementation checks and are not attributed to those sources.

References

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